SEOUL BIKE

HOURLY DEMAND

Will Seoul bikes be in High or Low demand this hour?

Supervised classification of hourly bike-sharing demand from weather and calendar data

95.6%

Random Forest accuracy on 1,752 unseen hours

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THE GOOD THE BAD THE LODOR

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8,760 hours, 14 columns

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The Problem: High or Low?

What we are predicting and why it matters

High or Low Demand, Hour by Hour

- Supervised binary classification: High is 505 or more rentals in the hour, Low is under 505
- Inputs: weather, hour of day, season, holiday and service status
- Why it matters: operators can move bikes before a rush empties a station

8,760

hourly records, Dec 2017 to Nov 2018

505

rentals per hour: the median, our cutoff

50 / 50

4,380 Low and 4,380 High hours

The Dataset at a Glance

8,760 hours · 14 columns · one row per hour of the Seoul public bike-sharing system

Time (2)

Date
Hour (0 to 23)

Weather (8)

Temperature
Humidity
Wind speed
Visibility
Dew point
Solar radiation
Rainfall
Snowfall

Calendar (3)

Seasons
Holiday
Functioning Day

TARGET **Rented Bike Count (0 to 3,556 per hour) becomes the High / Low label**

Source: Sathishkumar & Cho (2020), Seoul Bike Sharing Demand, UCI Machine Learning Repository

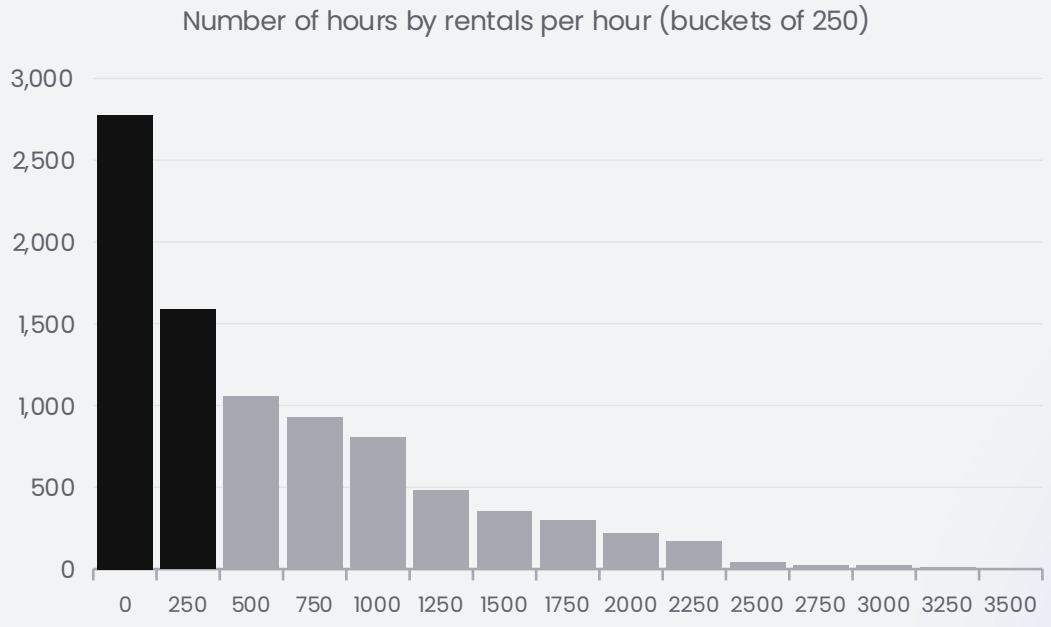
Data Cleaning Log

Already clean: columns renamed and types converted. No rows deleted, no values changed.

CHECK	Before	After
Column names	Spaces, brackets and ° symbols	Clean snake_case names
Date column	Text (str)	datetime64
Missing values	0	0 (nothing to fix)
Duplicate rows	0	0 (nothing to drop)
Closed hours	295 hours, all 0 rentals	**Kept: closed means Low**
Rows	8,760	8,760 (none removed)

FINDING 01

Demand Is Skewed



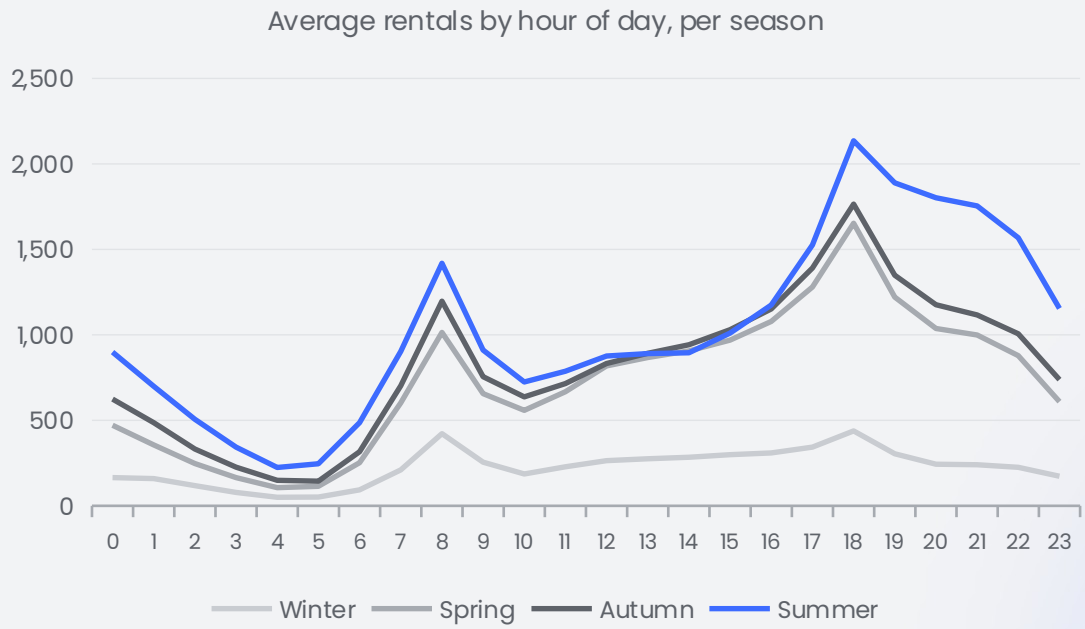
505

rentals per hour: the High / Low cutoff

- Mean 704.6, median 504.5
- Skewness 1.15, max 3,556
- The median cutoff gives a perfect 50 / 50 split
- Black bars: hours below the cutoff

FINDING 02

Two Daily Commute Peaks



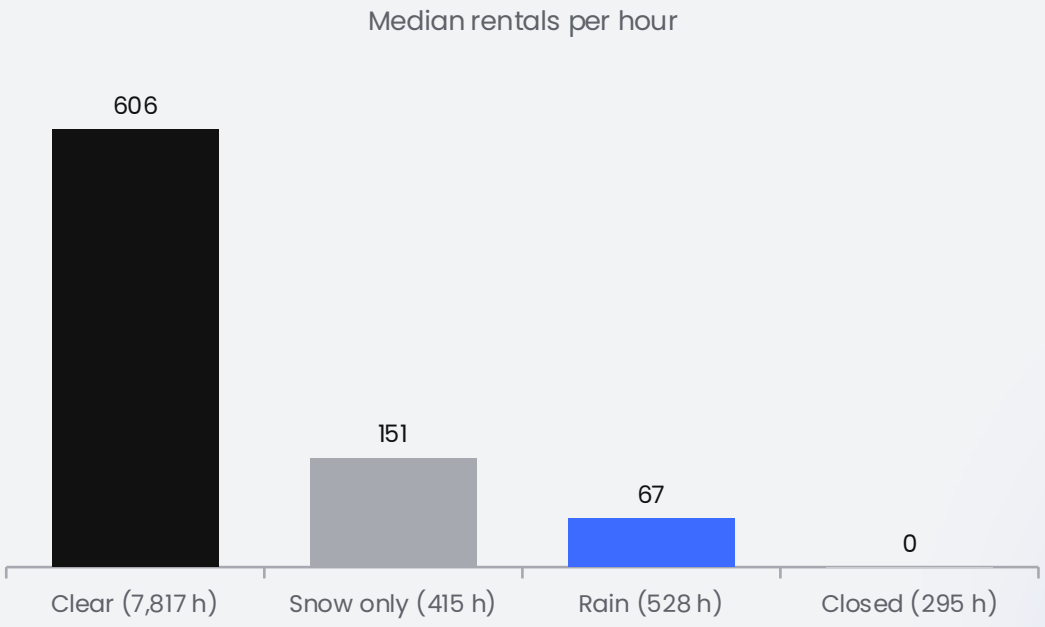
2,135

average rentals at 6 PM in summer

- 8 AM peak: about 1,016 bikes on average
- 6 PM peak: about 1,503 on average
- Winter never averages above 438 in any hour

FINDING 03

Rain and Closures Cut Demand



606 → 67

median rentals, clear vs. rain

- Rain cuts the median by about 90%
- Snow-only hours drop to 151
- All 295 closed hours have exactly 0 rentals

Feature Engineering

Four steps turn the raw table into model inputs

1

Label

Demand_Class = 1 if rentals ≥ 505, else 0

2

Calendar

Month and DayOfWeek extracted from Date

3

No Leakage

Rented Bike Count and Date dropped from X

4

Scale + Encode

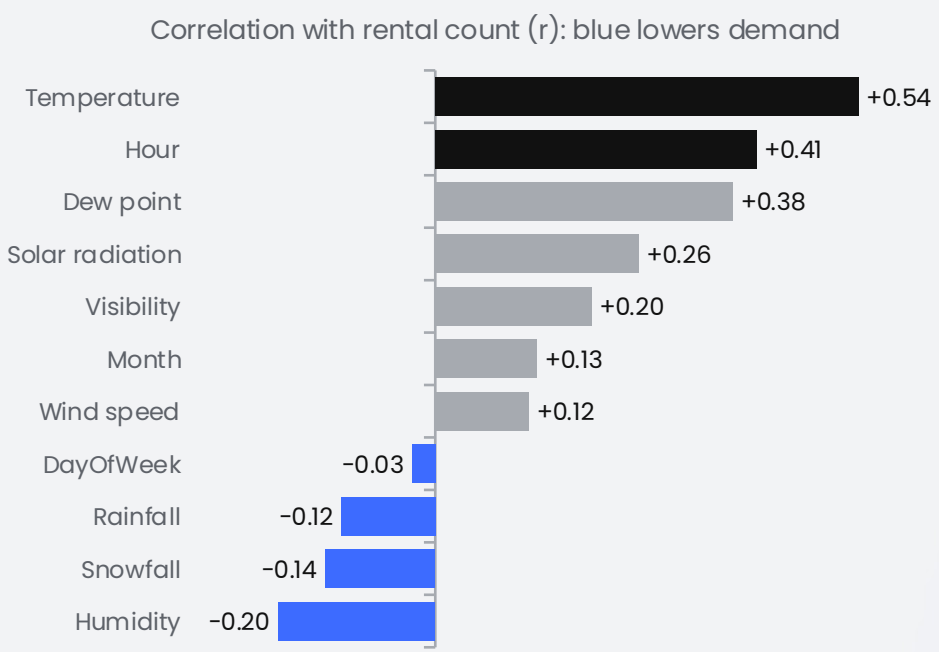
StandardScaler on 11 numeric, one-hot on 3 categorical

---

14 → 16 input columns become numeric predictors

Train only scaler and encoder never see test hours

Temperature and Hour Lead



0.913

temperature vs. dew point: near duplicates

Drop-Dew-Point Check

86.93% with **86.99%** without

Only one test hour apart, so every feature stays in.

Data Split and Models

7,008 / 1,752

train / test hours · 80-20 split · random_state=42

Preprocessing is learned from the training hours only, then applied to the test hours.

THREE MODELS, SIMPLEST TO MOST FLEXIBLE

Baseline

Always guesses the most common class

Logistic Regression

Linear, interpretable benchmark

Random Forest

100 trees, max depth 16; handles non-linear effects

Random Forest Wins at 95.6%

Accuracy on the 1,752 test hours

49.5%

Baseline

885 errors

86.9%

Logistic Regression

229 errors

95.6%

Random Forest

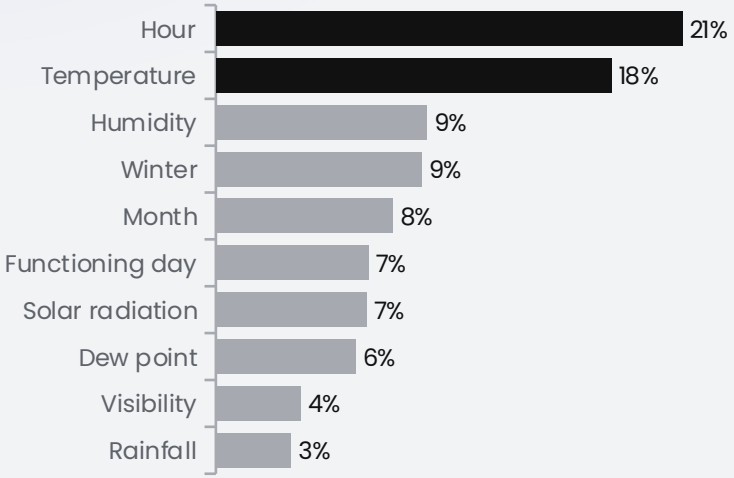
78 errors

Random Forest Confusion Matrix

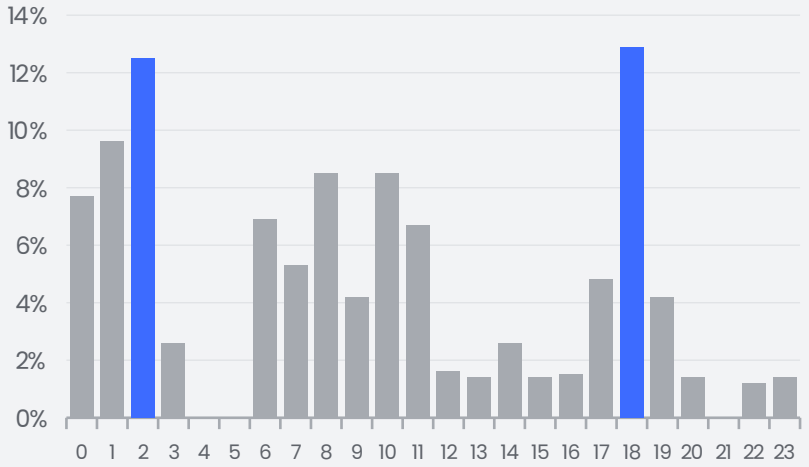
854 TN · Low, correct	**31** FP · false alarm
47 FN · missed rush	**820** TP · High, correct

Where Random Forest Goes Wrong

Top 10 feature importances



Error rate by hour of day (blue: hardest hours)



20.7% + 17.6%

Hour and temperature lead the model

54 of 78

errors sit within 150 rentals of the cutoff

3.5% to 5.1%

error rate in every season

Limits and Takeaways

LIMIT

Past-to-Future Split

Trained on the first 80% of the year, Random Forest drops to 85.1% (Logistic 86.1%)

TAKEAWAY

Weather and Time Predict It

Hour, temperature and weather alone reach 95.6% on unseen hours

LIMIT

City-Wide Totals Only

No station detail, and only 2017 to 2018 data from one city

TAKEAWAY

Errors Sit at the Edge

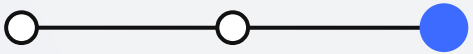
Random Forest beats Logistic Regression by 8.6 points; what it misses is near 505

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Thank You

Questions?

Seoul Bike Demand Classification · INTDSCI Final Project



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